

Exotic Spheres & Multiplicity Topology

0. Executive idea

We construct **prime-indexed cobordism invariants** for exotic spheres by translating surgery/handle data into **recursive multiplicity matrices** and then extracting **prime-tier spectral signatures** using multiplicity-style projectors. The guiding principle is:

Smooth structure obstructions (exoticness) should appear as *prime-layered defects* in a canonical multiplicity carrier, stable under cobordism moves.

This document organizes the proposal as a mathematical program and a fast validation plan.

1. Background: exotic spheres as smooth obstructions

A smooth **exotic sphere** is a manifold homeomorphic but not diffeomorphic to the standard sphere. In high dimensions ($n \geq 5$), exotic spheres are classified via surgery theory, stable homotopy, and bordism-type invariants (e.g., Kervaire–Milnor groups, Eells–Kuiper invariants in dimension 7, and refinements in modern classifications).

Multiplicity Theory angle: treat primes as identity/eigenmode labels; “multiplicity” encodes recursion depth/interaction thickness. The topological/smooth structure becomes a structured carrier that can be decomposed into prime tiers.

2. Goal

2.1 Prime-indexed cobordism invariants

Given a smooth homotopy sphere Σ^n , produce a **family of invariants**

$$\{\mathfrak{I}_{p^r}(\Sigma)\}_{p \in \text{prime}, r \geq 1}$$

that:

- is **invariant under bordism / surgery moves** after canonicalization,
- is **sensitive to smooth structure** (so it can distinguish exotic spheres), and
- is **computable** from presentations (plumbing graphs / Kirby diagrams / handle decompositions).

2.2 Physical implication

Use the prime spectrum $\{\mathfrak{I}_{p^r}\}$ to define weights/phases for a **sum over smooth structures** in quantum gravity, and to refine **string topology** operations by prime-tier projections.

3. The multiplicity bordism picture (high-level)

We model the bordism category as acting on multiplicity carriers:

- Objects: n -manifolds (e.g., Σ^n)
- Morphisms: bordisms $W : \Sigma_0 \rightsquigarrow \Sigma_1$
- Functorial target: a prime-graded algebra/category of multiplicity carriers

3.1 Multiplicity carrier

For each Σ choose presentation data (surgery or handles) and build:

- a hypergraph (or graph) $H(\Sigma) = (V, E, \iota)$
- a matrix/operator M_Σ built recursively from attaching/intersection/linking/framing data

The carrier is $(H(\Sigma), M_\Sigma)$.

4. Construction: recursive multiplicity matrix from surgery/handle data

4.1 Input formats

Any of the following can serve as the input presentation $\mathcal{P}(\Sigma)$:

- Kirby diagram: framed link in S^3
- Plumbing graph (common for 7-spheres and Brieskorn-type constructions)
- Morse handle decomposition

4.2 Data extraction

From $\mathcal{P}(\Sigma)$, extract a bilinear kernel K capturing **smooth-sensitive** structure:

- intersection form / linking matrix (depending on dimension and encoding)
- framing defects
- characteristic class refinements (spin/pontryagin corrections)
- optionally spectral/eta-type corrections for smooth sensitivity

4.3 Prime labeling and recursion depth

Assign to each generator/handle/node $v \in V$:

- prime label $p(v)$
- multiplicity exponent $\mu(v) \in \mathbb{N}$ (recursion depth / interaction thickness)

A deterministic choice (for reproducibility): order nodes by some canonical rule and map them to the first $|V|$ primes; $\mu(v)$ can be derived from local graph depth or from handle index stratification.

4.4 Matrix definition

Define

$$M_{\{\Sigma\}}(i,j) := p(v_i)^{\mu(v_i)}, p(v_j)^{\mu(v_j)}, K(i,j).$$

Interpretation:

- $K(i, j)$ encodes topological attachment;
- prime-powers encode “identity + depth”;
- the product encodes coupling strength across nodes.

4.5 Recursion rule (optional but powerful)

Introduce a recursion operator $\mathcal{R}$ that updates μ (and possibly K) under moves, e.g.

$$\mu_{t+1}(v) = \big(\mu_t(v), \{\mu_t(u) : u \sim v\}, \text{handle move type}\big).$$

This makes the invariant *dynamic* but still reducible to a canonical normal form.

5. Prime-tier projectors and invariants

5.1 Prime-tier projector

For each prime power p^r , define an idempotent projector Π_{p^r} acting on matrices/operators so that $\Pi_{p^r} M$ keeps only the p^r -tier information.

(Implementation idea: reduce matrix entries to p -adic valuation layers, or filter by prime divisibility structure, or use operator-algebraic projections.)

5.2 Spectral packages (families of invariants)

Define prime-indexed signatures using any combination of:

- traces of powers: $\frac{I^{\mathrm{tr}}(p^r)(\Sigma)}{M_{\{\Sigma\}}^k} \big|_{k=1}^K$
- determinant/zeta forms: $\frac{I^{\mathrm{det}}(p^r)(\Sigma; s)}{M_{\{\Sigma\}}(I-s, \Pi)} = \det(I-s, \Pi)$
- eigenvalue multiplicity spectra: $\frac{I^{\mathrm{spec}}(p^r)(\Sigma)}{M_{\{\Sigma\}} \text{ (as multiset)}}$

These are intended to form a stable “prime fingerprint” of smooth structure.

6. Cobordism invariance strategy: canonicalization (“gauge fixing”)

The central technical problem is **presentation dependence**: different Kirby/handle presentations represent the same manifold.

6.1 Move invariance via normal forms

Define a canonicalization procedure $\mathcal{N}$ such that:

- if $\mathcal{P}$ and $\mathcal{P}'$ differ by Kirby/handle moves, then
$$\mathcal{N}(M_{\Sigma}(\mathcal{P})) = \mathcal{N}(M_{\Sigma}(\mathcal{P}'))$$

6.2 What canonicalization might look like

- reduce by elementary transformations encoding handle slides (row/column operations)
- eliminate canceling handle pairs (block reduction)
- normalize framings / blowups (stabilization moves)
- choose a canonical prime labeling up to symmetries (graph automorphisms)

In practice: treat moves as an action of a generated equivalence relation on M , and reduce to a canonical representative.

7. Exotic sphere sensitivity: where smoothness enters

To distinguish exotic spheres, K must incorporate **smooth-sensitive** refinements.

Candidate sources of smooth sensitivity:

- Eells–Kuiper-type invariants in dimension 7 (and generalizations)
- eta invariants / APS boundary correction terms
- stable framing defects and the J -homomorphism context
- refined characteristic class data (spin/pontryagin corrections)

The matrix framework does not replace these; it reorganizes them into a **prime-tier spectral signal**.

8. Implications for quantum gravity

8.1 Sum over smooth structures

If a fixed topological manifold admits multiple smoothings, define a “smoothness sector” partition contribution

$$Z_{\mathrm{smooth}}(M) := \sum_{\sigma \in \mathrm{Smooth}(M)} \exp\big(i \int_M S_{\mathrm{mult}}(\sigma)\big)$$

where S_{mult} is built from prime-tier invariants:

$$S_{\text{mult}}(\sigma) = \sum_{p,r} w_{p^r} \Phi\left(\frac{I_{p^r}(\sigma)}{\text{big}}\right)$$

for suitable weights w_{p^r} and phase map Φ .

Expected outcome: *coherent clustering or cancellation* across smooth sectors due to prime-tier resonance.

8.2 Noncommutative/operator viewpoint

Prime-tier operators naturally resemble noncommutative observables; canonicalization resembles gauge reduction. This aligns well with approaches that link exotic smoothness to operator-algebraic structures.

9. Implications for string topology

String topology builds algebraic operations on $H_*(LM)$ (free loop space). The proposal:

- enrich operations by prime-tier projectors Π_{p^r}
- interpret loop concatenation/splitting as interactions whose channels are filtered by prime structure

Potential deliverable: a **prime-decomposed BV algebra** on loop homology, where exotic smooth structure modifies the prime spectrum of operations.

10. Concrete predictions (testable)

1. **Dimension 7 clustering:** Milnor 7-spheres should split into distinguishable clusters by $\{\mathfrak{I}_{p^r}\}$, reflecting prime-primary components implicit in the classification groups.
 2. **Move stability:** applying Kirby/handle moves to the same sphere should leave $\mathfrak{I}_{p^r}$ unchanged *after canonicalization*.
 3. **Smooth sensitivity:** at least one small-prime tier (often $p = 2$ or 7 in 7D contexts) should systematically detect changes invisible to purely homotopy-level invariants.
 4. **Physics signature:** in a toy path-integral weighting, prime tiers may produce structured interference (selection of “dominant smoothings”).
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11. Fastest path to validation (MVE)

Step A — choose a small battleground

Use **7-dimensional exotic spheres** presented by explicit plumbing/link data.

Step B — implement pipeline

1. Parse a presentation $\mathcal{P}(\Sigma)$ (plumbing graph or Kirby diagram).
2. Build K (linking/intersection/framing form + smooth refinements).
3. Assign primes $p(v)$ and compute $\mu(v)$.
4. Construct M_Σ .
5. Apply Π_{p^r} for a few tiers and compute $\mathfrak{I}_{p^r}$.

Step C — invariance checks

- apply known Kirby moves / handle slides / blowups
- confirm invariants are stable after canonicalization

Step D — exotic discrimination

- compare two non-diffeomorphic spheres with known classification differences
- check whether prime spectra separate them

12. Reference map

12.1 Internal Multiplicity documents (project context)

- *Multiplicity Operator Calculus* (MOC)
- *Polymorphic Multiplicity Matrices*
- *Multiplicity Theory V0*

12.2 Exotic spheres, surgery, invariants:

- M. A. Kervaire & J. W. Milnor, “**Groups of Homotopy Spheres: I**”, *Annals of Mathematics* **77** (1963), 504–537.
- J. Eells Jr. & N. H. Kuiper, “**An invariant for certain smooth manifolds**”, *Annali di Matematica Pura ed Applicata* **60** (1962), 93–110. DOI: 10.1007/BF02412768.
- D. Crowley, “**The classification of 2-connected 7-manifolds**” (and published version in *Proceedings of the London Mathematical Society*). arXiv:1406.2226.
- E. V. Brieskorn, “**Examples of singular normal complex spaces which are topological manifolds**”, *PNAS* **55** (1966), 1395–1397. DOI: 10.1073/pnas.55.6.1395.
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- M. A. Hill, M. J. Hopkins, D. C. Ravenel, “**A solution to the Arf-Kervaire invariant problem**” (background/outline).

12.3 String topology

- M. Chas & D. Sullivan, “**String Topology**” (1999). arXiv:math/9911159.
- D. Sullivan, “**String topology: background and present state**” (survey).

12.4 Exotic smoothness & quantum gravity

- T. Asselmeyer-Maluga, **“Smooth quantum gravity: Exotic smoothness and Quantum gravity”** (2016). arXiv:1601.06436.
- T. Asselmeyer-Maluga & J. Król, **“Exotic Smoothness and Quantum Gravity II: exotic $\mathbb{R}^4$ $\mathbb{R}^4$, singularities and cosmology”** (2011). arXiv:1112.4882.

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